ENSTA Paris, ENSAE, **Institut Polytechnique de Paris**
Master Finalité SFA et ENSTA-ENSAE (3e année)

(Elements of Stochastic Calculus, FR) 2021-22

EXERCISES

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space and a filtration $(\mathcal{F}_t)$ fulfilling the usual conditions. Let W be an $(\mathcal{F}_t)$-Brownian motion.

25. Let (S_t) be a $BESQ(x_0, \delta)$. For $N > 0$, we set $\tau_N = \inf\{s \geq 0 | S_s \geq N\}$.

 (i) Show that a.s. (τ_N) is an increasing sequence of stopping times diverging to infinity.

 Solution.

 - τ_N is a stopping time by Proposition 4.11 for every N.
 - The sequence (τ_n) is obviously increasing.
 - For every $\omega \in \Omega$, we denote

 $$\tau(\omega) := \lim_{N \to \infty} \tau_N(\omega).$$

 We can show that $\tau(\omega) = \infty$ a.s.
 Indeed, if for some ω, $\tau(\omega) < \infty$ then for every N

 $$\sup_{t \leq \tau+1} S_t \geq \sup_{t \leq \tau_N + 1} S_t \geq N.$$

This would imply that on the compact $[0, \tau_N + 1]$ the supremum of the continuous function $t \mapsto S_t(\omega)$ is $+\infty$, which is impossible.

(ii) Show that, whenever N is large enough

$$E(S_t^{\tau_N}) = x_0 + \delta E(\tau_N \wedge t).$$

Solution. We have

$$S_t^{\tau_N} = x_0 + (\tau_N \wedge t) + M_t^N, \ t \in [0,T], \qquad (1)$$

and

$$M_t^N := 2\int_0^{\tau_N \wedge t} \sqrt{S_s} dW_s, \ t \in [0,T],$$

is a square integrable martingale. Indeed,

$$E\left(\int_0^T 1_{[0,\tau_N \wedge T]} S_s ds\right) \leq NT.$$

So, taking the expectation in (1) we get the result.

(iii) Calculate $E(S_t), t \geq 0$.

Solution. By Fatou's lemma, for every $t > 0$

$$\begin{aligned} E(S_t) &= E\left(\lim_{N\to+\infty} S_t^{\tau_N}\right) = E\left(\liminf_{N\to+\infty} S_t^{\tau_N}\right) \\ &\leq \liminf_{N\to+\infty} E(S_t^{\tau_N}) = \liminf_{N\to+\infty}(x_0 + \delta(\tau_N \wedge t)) \\ &= x_0 + \delta t. \end{aligned}$$

Now

$$E\left(\int_0^T S_s ds\right) < +\infty,$$

so

$$M_t := 2\int_0^t \sqrt{S_s} dW_s, t \geq 0,$$

is a (square integrable) martingale. Taking the expectation in

$$S_t = x_0 + \delta t + M_t,$$

we obtain

$$E(S_t) = x_0 + \delta t.$$

We take now $t = T$.

(iv) Show that $E(\sup_{t \le T} S_t^2) < \infty$.

Solutions.

We have

$$\sup_{t \le T} S_t^2 \le 3x_0^2 + 3\delta^2 T^2 + 3 \sup_{0 \le t \le T} M_t^2.$$

Taking the expectation and using Doob's, we get

$$E(\sup_{t \le T} S_t^2) \le 3x_0^2 + 3\delta^2 T^2 + 12E(M_T^2).$$

Now

$$\begin{aligned} E(M_t^2) &= 4\int_0^t E(S_s)ds = 4\int_0^t (x_0 + \delta s)ds \\ &= 4x_0 t + 2\delta t^2. \end{aligned}$$

(v) Calculate $\mathrm{Var}(S_t)$ for any $t \geq 0$.

Solutions.

$$\begin{aligned}\mathrm{Var}(S_t) &= E(S_t - x_0 - \delta t)^2 = E(M_t^2) \\ &= 4E\left(\int_0^t S_s ds\right) = 4x_0 t + 2\delta t^2.\end{aligned}$$

26. Deduce $E(S_t)$ if S is a Square Bessel process using Theorem 14.12.

Solutions. We set

$$\psi(\lambda) := \psi_{S_t}(\lambda) = E(\exp(-\lambda S_t)),$$

which gives

$$\psi'(0) = -E(S_t).$$

Since

$$\psi(\lambda) = \left(\frac{1}{2t\lambda+1}\right)^{\frac{\delta}{2}} \exp\left(-\frac{\lambda x}{2t\lambda+1}\right),$$

we get

$$\begin{aligned}
\psi'(\lambda) &= \left\{(2t\lambda+1)^{-\frac{\delta}{2}-1}\left(-\frac{\delta}{2}\right)2t\right. \\
&+ \left.(2t\lambda+1)^{-\frac{\delta}{2}}\frac{-x(2t\lambda+1)+\lambda x 2t}{(2t\lambda+1)^2}\right\} \\
&\times \exp\left(-\frac{\lambda x}{2t\lambda+1}\right) \\
&= \exp\left(-\frac{\lambda x}{2t\lambda+1}\right) \\
&\times \left\{-\delta t(2t\lambda+1)^{-\frac{\delta}{2}-1} - x(2t\lambda+1)^{-\frac{\delta}{2}-2}\right\}.
\end{aligned}$$

So

$$\psi'(0) = -\delta t - x,$$

and

$$E(S_t) = \delta t + x.$$

EXERCISES

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space and a filtration $(\mathcal{F}_t)$ fulfilling the usual conditions. Let W be an $(\mathcal{F}_t)$-Brownian motion.

27. Let $S^i, i = 1, 2$ be the solution of

$$S_t^i = x_i + 2\int_0^t \sqrt{S_s^i} dW_s + \delta_i t, i = 1, 2,$$

where $0 \leq x_1 \leq x_2, 0 \leq \delta_1 \leq \delta_2$.

(a) Show that $S_t^1 \leq S_t^2, \forall t \in [0, T]$ in the sense of indistinguishability.

Solution.

This is a consequence of the comparison theorem.

(b) Express $E(|S_t^1 - S_t^2|)$.

Solution.

We have

$$\begin{aligned} E(|S_t^2 - S_t^1|) &= E(S_t^2 - S_t^1) = E(S_t^2) - E(S_t^1) \\ &= (x_2 - x_1) + (\delta_2 - \delta_1)t, \end{aligned}$$

see Exercise 25.

(c) Show that

$$\mathrm{Var}(S_t^2 - S_t^1) \leq 4(x_2 - x_1)t + 2(\delta_2 - \delta_1)t^2.$$

Solution.

$$\begin{aligned}
\mathrm{Var}(S_t^2 - S_t^1) &= 4E\left(\int_0^t (\sqrt{S_s^2} - \sqrt{S_s^1})dW_s\right)^2 \qquad (1)\\
&= 4E\left(\int_0^t (\sqrt{S_s^2} - \sqrt{S_s^1})^2 ds\right)\\
&\leq 4E\left(\int_0^t |S_s^2 - S_s^1| ds\right),
\end{aligned}$$

since

$$|a-b|^2 \leq |a^2 - b^2|, \quad a, b \geq 0.$$

So previous expectation gives

$$\begin{aligned}
E\left(\int_0^t (S_s^2 - S_s^1)ds\right) &= \int_0^t E(S_s^2 - S_s^1)ds\\
&= \int_0^t ((x_2 - x_1) + (\delta_2 - \delta_1)s)ds\\
&= (x_2 - x_1)t + (\delta_2 - \delta_1)\frac{t^2}{2}
\end{aligned}$$

(d) Using Doob's inequality find an upper bound for

$$E\left(\sup_{t\le T}|S_t^2 - S_t^1|^2\right).$$

Solution.

We have

$$\begin{aligned} S_t^2 - S_t^1 &= x_2 - x_1 + (\delta_2 - \delta_1)t \\ &+ 2\left(\int_0^t (\sqrt{S_s^2} - \sqrt{S_s^1})dW_s\right). \end{aligned} \tag{2}$$

So

$$E\left(\sup_{t\le T}|S_t^2 - S_t^1|^2\right) \le 3(x_2 - x_1)^2 + 3(\delta_2 - \delta_1)^2 T^2$$
$$+\ 12E\left(\sup_{t\le T}\left(\int_0^t (\sqrt{S_s^2} - \sqrt{S_s^1})dW_s\right)^2\right).$$

By Doob's, this is smaller or equal than

$$\begin{aligned}
3(x_2 - x_1)^2 &+ 3(\delta_2 - \delta_1)^2 T^2 \\
&+ 12 \times 4E\left(\int_0^T (\sqrt{S_s^2} - \sqrt{S_s^1})dW_s\right)^2 \\
&= 3(x_2 - x_1)^2 + 3(\delta_2 - \delta_1)^2 \\
&+ 12 \times 4\frac{Var(S_T^2 - S_T^1)}{4},
\end{aligned}$$

by (1). By (c), this is smaller than

$$\begin{aligned}
3(x_2 - x_1)^2 &+ 3(\delta_2 - \delta_1)^2 T^2 \\
&+ 12\left(4(x_2 - x_1)T + 2(\delta_2 - \delta_1)T^2\right) \\
&= 3(x_2 - x_1)^2 + 3(\delta_2 - \delta_1)^2 T^2 \\
&+ 48(x_2 - x_1)T + 24(\delta_2 - \delta_1)T^2.
\end{aligned}$$

(e) Let (x_n) (resp. δ_n) be an increasing sequence of non-negative numbers converging to x_0 (resp. δ_0). Let $S = S^n$ the solution to

$$S_t = x_n + 2\int_0^t \sqrt{S_s}dW_s + \delta_n t, \quad n \in \mathbb{N}.$$

Deduce that the sequence S^n converges ucp to S^0.

Solution.

By previous estimates (d)

$$\begin{aligned} E\left(\sup_{t\leq T}|S_t^n - S_t^0|^2\right) &\leq 3(x_0 - x_n)^2 + 3(\delta_0 - \delta_n)^2T^2 \\ &+ 48(x_0 - x_n)T + 24(\delta_0 - \delta_n)T^2 \\ &\quad \to_{n\to+\infty} 0. \end{aligned}$$

So the sequence S^n converges ucp to S^0, since the convergence in L^2 implies the convergence in probability.

(f) Let us adopt the notations of Chapter 14. Deduce that $\varphi(x_n, \delta_n) \to \varphi(x, \delta)$ in the case when μ has its support on a compact interval $[0, T]$.
Solution.

We have

$$\varphi(x_n, \delta_n) = E\left(\exp\left(-\lambda \int_0^T S_s^n d\mu(s)\right)\right).$$

Let (n_k) be a sequence diverging to $+\infty$. it is enough to show the existence of a subsequence (n_{k_j}) such that

$$\varphi(x_{n_{kj}}, \delta_{n_{kj}}) \to \varphi(x_0, \delta_0).$$

By (e) there is a sequence (n_{k_j}) such that

$$\sup_{t \le T} |S_t^{n_{kj}} - S_t^0| \to_{j\to\infty} 0,$$

a.s. By Lebesgue's dominated convergence theorem the result follows.

28. Let X be a Cox-Ingersoll-Ross process. Determine the Laplace trasform of the law of X_t i.e. $E(e^{-\lambda X_t})$, for $\lambda > 0$.

Solutions.

By Proposition 15.5, if (T_t) is a BESQ$(x_0, \delta))$ with $x_0 = \frac{c^2}{b}$, then

$$X_t = e^{bt} T_{\varphi(t)},$$

with

$$\varphi(t) = \frac{c^2}{4b}(1 - e^{-bt}),$$

is a Cox-Ingersoll-Ross process. We have

$$\begin{aligned} E(e^{-\lambda X_t}) &= E(\exp(-\lambda e^{bt} T_{\varphi(t)}) \\ &= E(\exp(-\tilde{\lambda} T_{\varphi(t)})), \end{aligned}$$

with $\tilde{\lambda} = \lambda e^{bt}$. By Theorem 14.12 we get

$$
\begin{aligned}
E(\exp(-\tilde{\lambda} T_{\varphi(t)})) &= \left(\frac{1}{2\varphi(t)\tilde{\lambda}+1}\right)^{\frac{\delta}{2}} \exp\left(-\frac{\tilde{\lambda} x_0}{2\varphi(t)\tilde{\lambda}+1}\right) \\
&= \left(\frac{1}{2\lambda e^{bt}\frac{c^2}{4b}(1-e^{-bt})+1}\right)^{\frac{2a}{c^2}} \\
&\quad \times \exp\left(-\frac{\lambda e^{bt}\frac{c^2}{b}}{\frac{c^2}{2b}(1-e^{-bt})\lambda e^{bt}+1}\right),
\end{aligned}
$$

which equals

$$
\left(\frac{2b}{\lambda c^2(e^{bt}-1)+2b}\right)^{\frac{2a}{c^2}} \exp\left(-\frac{\lambda e^{bt}\frac{c^2}{b}}{\frac{c^2}{2b}\lambda(e^{bt}-1)+1}\right).
$$